Song structure in a 10-hour humpback recording (LeiData 193500)

Summary

- Song cycle of about 12.8 minutes. Long motifs recur all night at a steady interval, consistent with a humpback song of about 13 minutes. The 75-minute recording 011115+014845 gave a similar cycle (about 14 minutes).
- Stable across motif lengths. From 6 to 10 minutes, neighbouring lengths find nearly the same number of occurrences (47–54), the same start-to-start intervals and mostly the same positions. The 2-, 4-, 8- and 16-minute lengths do not.
- 1-minute phrases with fixed places. Seven groups of 1-minute phrases each sit at their own place in the song.
- A shared order. 27 of the 42 songs that contain at least two of the phrase groups follow the order G7 → G6 → G3 → G1 → G5 → G2 → G4 exactly. With the order inside each song shuffled, the best order never matched more than 18 songs ($p < 0.01$).
- Energy minima fall inside the song marker, not at the song boundary. Clear minima of the slowly varying energy occur in about every other song. They sit about 2 minutes before the centre of the 7-minute marker, at relative location about 0.35, where the first phrase of the linear order (G7) also sits.

Data and method

File	`LeiData/SOCAL_H_76_230426_193500_df100.x.x.wav`, starting 2023-04-26 19:35:00
Audio	10 h, mono, 2 kHz (decimated from 200 kHz, so content up to 1 kHz)
Features	MFCC coefficient 2 at 10 frames/s (0.1 s window and hop), 360,000 frames

1. Matrix profiles (`profiles193500.m`, `whaleComputeProfiles`).
   - For each motif length, `mpxForPan` finds every window's nearest non-overlapping match. The correlation of the match is $r = 1 - \mathrm{mp}^2/(2w)$.
   - Lengths: 1, 2, 4, 6 and 7–20 minutes, saved once in `runs/193500/profiles_c2.mat`. Each length takes about 3 minutes.
2. One motif group per length (`motifLengths193500.m`).
   - Seed: the best pair, which must have r ≥ 0.2.
   - Copies: windows whose correlation with the seed reaches a copy threshold, overlapping by at most 10%. Copies therefore start at least 0.9 × the length apart.
   - Copy threshold: it is lowered from 0.2 until no stretch of more than 20 minutes is left without a copy.
3. Start-to-start intervals. Neighbouring lengths are compared by the number of occurrences and by two distances between their interval distributions:
   - KS D: the largest gap between the cumulative distributions (0 means identical).
   - W: the average shift between the distributions, in minutes.

Song cycle

The best-match correlation falls as the window grows, because a longer window includes more non-matching song. Choosing the length with the most correlated pair therefore always picks the shortest length allowed.

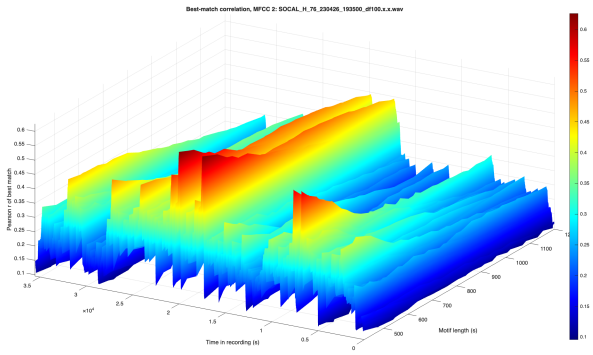


Figure 1: Best-match correlation, 7–20 min

Length (min)	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Best r	0.63	0.60	0.57	0.55	0.52	0.50	0.50	0.51	0.50	0.50	0.49	0.47	0.46	0.44

Motif length scan

The aim is two neighbouring lengths with the same number of occurrences and similar start-to-start histograms. A part of the song that occurs once per song should behave the same at any two lengths that fit inside it.

Coarse scan: 2, 4, 8 and 16 minutes. No pair matches.

Pair	Occurrences	KS D	W
2 vs 4 min	125 vs 66	0.62	4.5 min
4 vs 8 min	66 vs 51	0.36	2.6 min
8 vs 16 min	51 vs 31	0.90	7.4 min

- Short motifs repeat within a song. 2- and 4-minute motifs recur several times per song (median intervals 3.8 and 8.5 minutes).
- 16 minutes is longer than a song. Its copies cannot recur every cycle: the copy threshold fell to −1 without closing every 20-minute stretch.

Fine scan: 6–10 minutes. Every neighbouring pair matches.

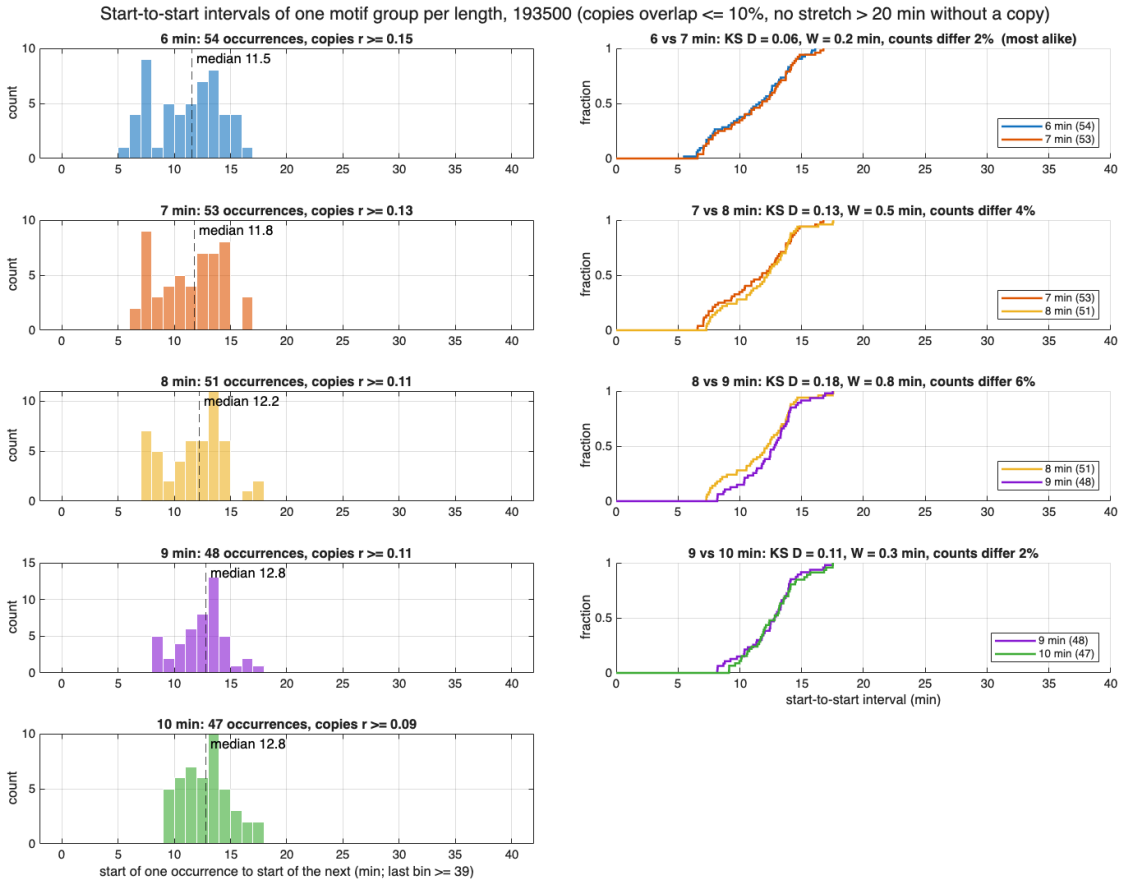


Figure 2: Start-to-start intervals, 6–10 min

Length	Occurrences	Copy threshold	Median interval (IQR)	Intervals < 10 min
6 min	54	0.15	11.5 (8.0–13.7)	20 of 53
7 min	53	0.13	11.8 (8.5–13.7)	18 of 52
8 min	51	0.11	12.2 (9.7–13.8)	14 of 50
9 min	48	0.11	12.8 (11.1–13.9)	7 of 47
10 min	47	0.09	12.8 (11.4–14.0)	5 of 46

“Same occurrences” in the next table counts the occurrences of the shorter length whose midpoint lies within 1 minute of an occurrence of the longer length.

Pair	Count difference	KS D	W	Same occurrences
6 vs 7	2%	0.06	0.2 min	51 of 54
7 vs 8	4%	0.13	0.5 min	47 of 53
8 vs 9	6%	0.18	0.8 min	44 of 51
9 vs 10	2%	0.11	0.3 min	44 of 48

- One set of positions. Across 6–10 minutes the method finds one stable set of positions, about one per song.
- 6 and 7 minutes match most closely, but about 38% of their intervals are under 10 minutes. These are half-song matches: the motif matched twice within one song.
- 9 and 10 minutes give the cleanest cycle. Both have a median interval of 12.8 minutes and a single peak at 13–14 minutes. A window longer than half a song can no longer match both halves.

Song markers. The analyses below use the 7-minute group as the song marker: 53 occurrences with midpoints M_k (`motifsVarLen193500.m`). Some of these markers are half-song matches. A 9–10-minute marker would be cleaner.

Inside the song: 1-minute phrases

Definitions.

- Songs are bounded halfway between consecutive markers, $B_k = (M_k + M_{k+1})/2$. This gives 51 songs, with a median length of 12.0 minutes (range 7.1–15.7).
- Relative location of a phrase with midpoint t in song k is $(t - B_k)/(B_{k+1} - B_k)$: 0 is the start of the song and 1 the end.

Phrase groups (`songRelLocation193500.m`).

- Seeds and copies need r ≥ 0.35, with at most 30 copies per group and no overlaps.
- The search found 29 groups. The 7 groups with at least 8 occurrences are kept as G1–G7; the other 22 have only 2–3 occurrences each.
- The 7 kept groups cover 2.8 h (28%) of the recording; all 29 groups cover 3.7 h (37%).

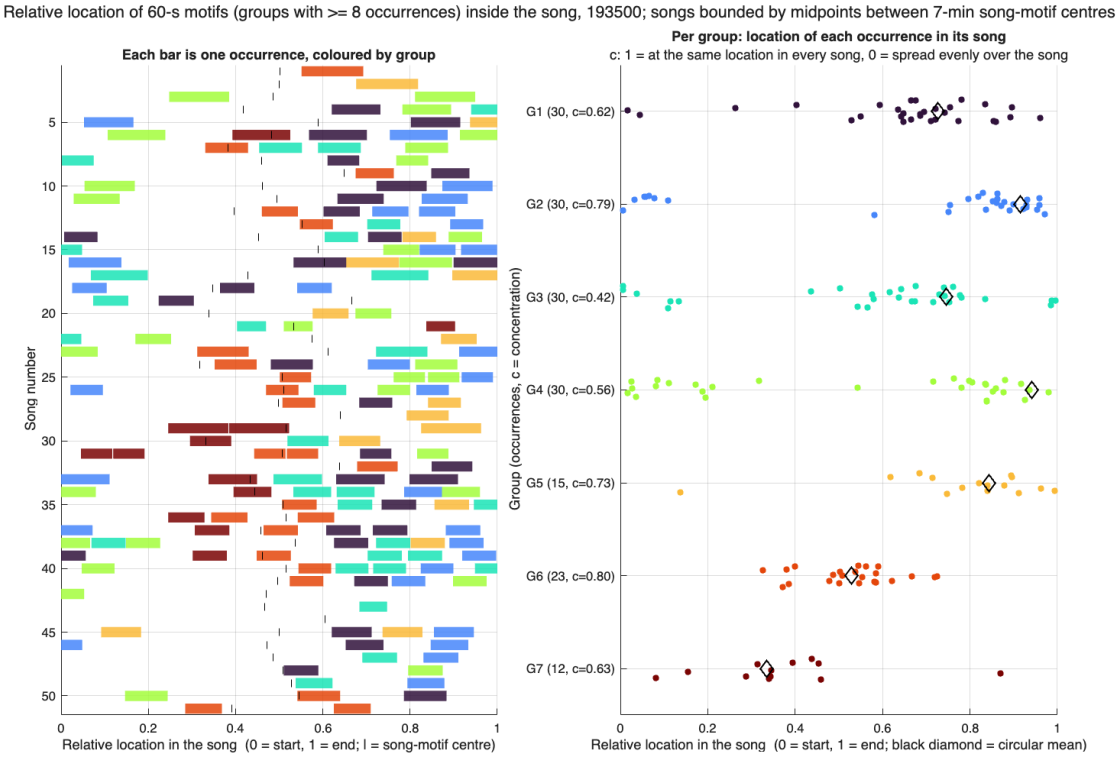


Figure 3: Relative location of the 1-minute phrases

In the figure:

- Left: one row per song, with each occurrence as a bar coloured by group.
- Right: one row per group. The diamond is the circular mean, an average that treats 0 and 1 as the same point.
- Concentration c measures how tightly a group clusters: 1 means at the same place in every song, 0 means spread evenly.

Group	Occurrences	Songs	Seed r	Circular mean	c
G1	30	26	0.78	0.73	0.62
G2	30	24	0.71	0.91	0.79
G3	30	22	0.70	0.75	0.42
G4	30	25	0.66	0.94	0.56
G5	15	14	0.59	0.84	0.73
G6	23	20	0.54	0.53	0.80
G7	12	10	0.50	0.33	0.63

- Each group has its own place. G7 and G6 come early (0.3–0.55), G1 and G3 at about 0.7, and G5, G2 and G4 at the end.

- The end wraps around. The occurrences of G2 and G4 near 0 belong to the end-of-song cluster, because the midpoint boundaries cut through it.

Order of the phrases

`songOrder193500.m` lists the kept groups of each song in time order, with consecutive repeats merged. The song start is moved to the largest empty stretch of relative locations (0.24), so that the end-of-song run stays in one song.

Pairwise order. These pairs have a strong order within songs:

Earlier → later	In this order	p
G6 → G2	12 of 12	0.0005
G1 → G4	15 of 16	0.0005
G6 → G1	11 of 11	0.001
G6 → G3	11 of 11	0.001
G1 → G2	13 of 15	0.007

Pairs within one song are not independent, so these p-values are only indicative.

Consensus order. The script looks for the order of all 7 groups that the most songs agree with, ignoring groups missing from a song.

- Linear: a song agrees if its groups appear in exactly this order from the song start.
- Circular: a song agrees if its groups follow the cycle, starting anywhere.
- Significance: the order inside each song is shuffled 100 times and the search is repeated on each shuffled data set.

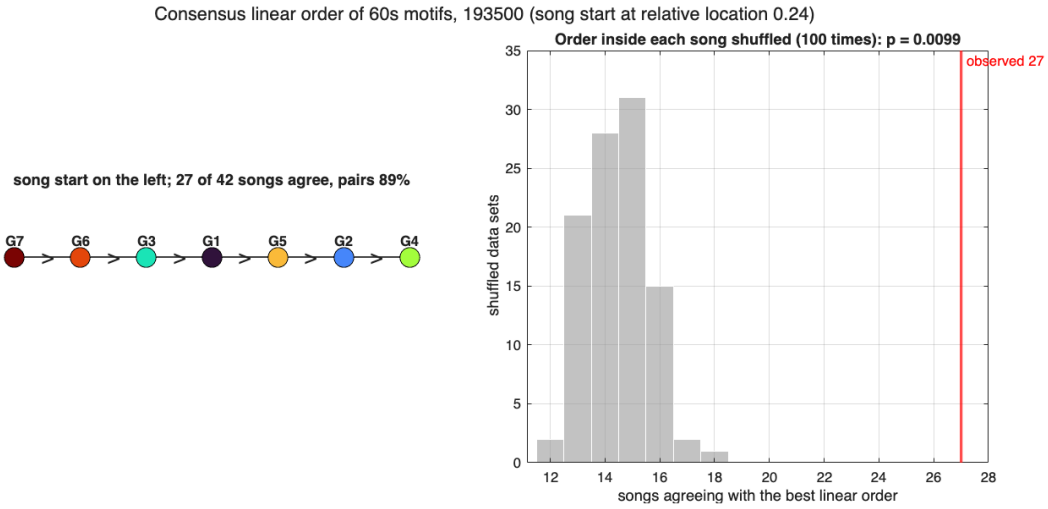


Figure 4: Consensus linear order and shuffle test

	Linear	Circular
Best order	G7 → G6 → G3 → G1 → G5 → G2 → G4	G6 → G3 → G1 → G5 → G2 → G4 → G7 → (G6)
Songs that agree	27 of 42	21 of 32
Fixed random order	7.7	9.2
Best order on shuffled songs	mean 14.5, max 18	mean 15.4, max 19
p	< 0.01	< 0.01

- The order is real. Both orders are far above chance. The p-value is the smallest that 100 shuffles can give.
- The two orders agree. The cycle is the linear order with G7, the rarest group, moved from the start to the end.
- G3 causes most disagreements. It appears in 12 of the 15 songs that disagree, often both early and late in the same song, and has the lowest concentration. It may be two different phrases matched by one template.

Energy minima and the song boundary

The question is whether the quietest parts of the recording mark the breaks between songs, and where they fall relative to the 7-minute song marker.

Method (`bandEnergy193500.m`, then `minimaRelLocation193500.m`).

- Energy: MFCC coefficient 0, the overall energy below 1 kHz.
- Slow energy curve: the 30-minute running mean is subtracted, and the result is smoothed with a Gaussian of 20-minute window (standard deviation 4 minutes).
- Minima: local minima of this curve with prominence at least 0.01 and at least 8 minutes apart. Each is placed in its song, using the same boundaries and relative location as for the phrases.
- Rayleigh test: tests whether the relative locations, treated as points on a circle, are spread evenly. R is their concentration (0 = even, 1 = all at one place).

The prominence threshold matters. Subtracting the running mean leaves a ripple of about 10^{-6} on the curve, with thousands of tiny minima. Without the threshold, 47 minima were reported:

- 15 were ripple on peaks or slopes of the curve (prominence below 10^{-4});
- 5 were shallow dips (prominence 7×10^{-4} to 3×10^{-3});
- the rest had prominence above 0.01.

With the threshold, the 8-minute separation no longer changes the result.

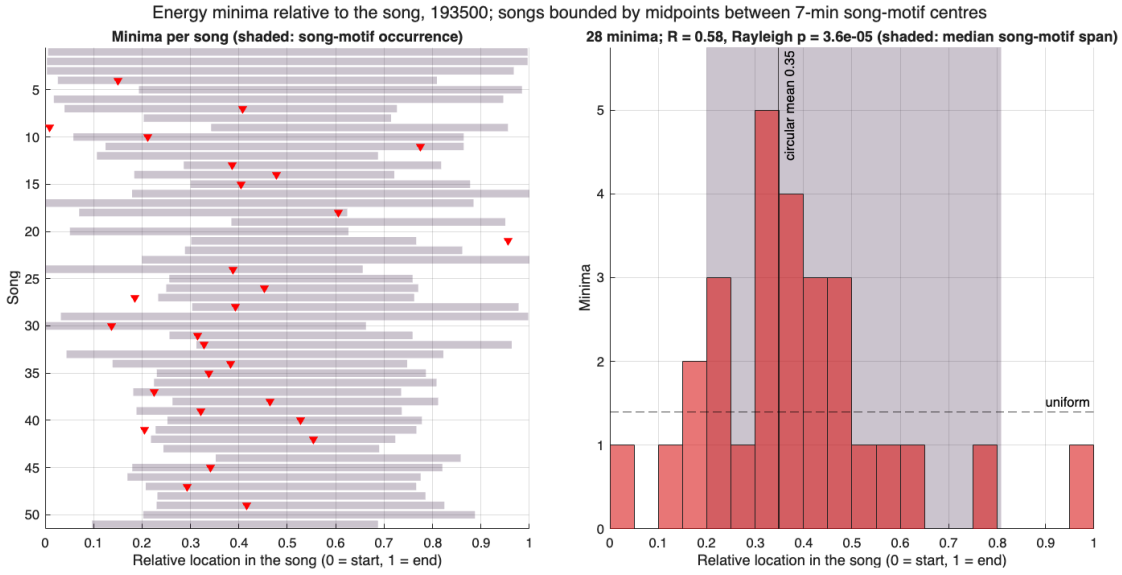


Figure 5: Energy minima inside the song

Minima	29, of which 28 lie inside a song
Interval between minima	median 16.9 min (IQR 13.4–25.6)
Songs with a minimum	28 of 51 (none has two)
Relative location	median 0.38 (IQR 0.28–0.46), circular mean 0.35
Concentration	R = 0.58, Rayleigh p = 4 × 10^{-5}
Inside a 7-min marker occurrence	25 of 29 (86%)
Offset from the nearest marker centre	median −1.9 min (IQR −2.4 to +0.2); 21 of 29 before the centre

- Not one per song. The minima are about 17 minutes apart, and only about half of the songs contain one. The slow energy is not a song-by-song marker at this smoothing.
- Not at the midpoint boundaries. Only 2 of the 28 minima lie within 0.1 of a boundary (relative location near 0 or 1).
- Inside the marker's first half. When a minimum occurs, it falls inside the 7-minute marker, about 2 minutes before its centre (relative location about 0.35).
- Same place as G7 and the order's start. G7, the first group of the linear order, has circular mean 0.33. The order analysis put the song start at 0.24. So the quietest part of the song is where the phrase order begins. If the energy

minimum marks the break between songs, a marker centred on the break is the current marker moved about 2 minutes (0.15 songs) earlier.

Caveats

- Weak copies. Capturing every song needs copy thresholds of 0.09–0.15, so many copies correlate only 0.1–0.3 with the seed. The regular spacing of the copies is the stronger evidence.
- Minimum interval. No start-to-start interval can be shorter than 0.9 × the motif length, and filling the long stretches pushes weak copies towards this minimum.
- Limited features. Matching uses one MFCC coefficient of audio below 1 kHz.
- Half-song markers. These distort the song boundaries and the relative locations.
- Greedy phrase groups. G1–G4 hit the 30-copy cap. Other settings give different groups, so labels are not comparable across runs.

Reproduce

```
profiles193500          % profiles for 7–20 min (once, about 35 min)
motifLengths193500      % length scan; set lensMin ([2 4 8 16] or [6 7 8 9 10])
motifsVarLen193500      % 7–min song markers and their midpoints
songRelLocation193500   % 1–min phrase groups, relative locations, coverage
songOrder193500         % order statistics and consensus orders
bandEnergy193500        % slow energy curve, its minima, detections
minimaRelLocation193500 % energy minima inside the songs
```

Settings are at the top of each script. Outputs, including CSVs and `.fig` files, are in `runs/193500/`.